

ESP-IDF Basic UART Example

This example configures UART1 at 115200 baud and transmits a message every second.

```
#include <stdio.h>
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "driver/uart.h"
#include "esp_log.h"

#define UART_PORT_NUM      UART_NUM_1
#define UART_TX_PIN        17
#define UART_RX_PIN        16
#define UART_BUF_SIZE      1024

static const char *TAG = "UART_EXAMPLE";

void app_main(void)
{
    uart_config_t uart_config = {
        .baud_rate = 115200,
        .data_bits = UART_DATA_8_BITS,
        .parity     = UART_PARITY_DISABLE,
        .stop_bits  = UART_STOP_BITS_1,
        .flow_ctrl  = UART_HW_FLOWCTRL_DISABLE,
    };

    uart_driver_install(UART_PORT_NUM,
                        UART_BUF_SIZE * 2,
                        0,
                        0,
                        NULL,
                        0);

    uart_param_config(UART_PORT_NUM, &uart_config);

    uart_set_pin(UART_PORT_NUM,
                 UART_TX_PIN,
                 UART_RX_PIN,
                 UART_PIN_NO_CHANGE,
                 UART_PIN_NO_CHANGE);

    const char *message = "Hello from ESP-IDF UART!\\r\\n";

    while (1)
    {
        uart_write_bytes(UART_PORT_NUM,
                         message,
                         strlen(message));

        ESP_LOGI(TAG, "UART Message Sent");

        vTaskDelay(pdMS_TO_TICKS(1000));
    }
}
```